

# MES HUMMER — SPESIFIKASI TEKNIS & FITUR APLIKASI

## Dokumen Acuan Penyusunan RAB (Rencana Anggaran Biaya)

Versi 1.0 · 2 September 2026 · Angka terverifikasi dari kode & database produksi

### 1. Ringkasan Eksekutif

- Aplikasi MES + WMS "Hummer" untuk manufaktur genteng/keramik: mencatat produksi dari start shift mesin MAKER, proses TSG (tanah sisa giling), HLP (packing batangan), rantai WR-SLOP-BAL, karton, dispatch, hingga laporan manajemen multi-plant.
- Skala saat ini: 66 tabel PostgreSQL, 133 file route API (REST /api/v1), 43 halaman UI, 14 role dengan 72 permission, 25 modul service, 34 migrasi DB, 10 spec E2E, CI/CD GitHub Actions 5 job.
- Dua aplikasi klien: (1) Web + Tablet (Next.js 15), (2) Mobile Android/iOS (Flutter, repo terpisah) dengan FCM push dan deep-link.
- Status: Fase 0-6 selesai, live di produksi (Vercel + Neon PostgreSQL + Coolify Docker). Sisa: checklist go-live lapangan dan pengembangan aspirasional.

### 2. Arsitektur Sistem

- Klien: Web admin (Next.js App Router), Tablet lantai produksi (web responsive), Mobile app (Flutter, REST + FCM).
- Backend: REST API /api/v1 (API-first, bukan Server Actions) -> layer service (25 modul) -> Drizzle ORM -> PostgreSQL 16.
- Keamanan berlapis: JWT (jose) + OTP 2 lapis + RBAC (middleware tiap route) + Row Level Security PostgreSQL per plant.
- Kalkulasi bisnis 100% server-side (yield, berat/batang, pembagian proporsional, OEE).
- Background: instrumentation auto-cleanup sesi (24 jam), push FCM fire-and-forget, soft delete + audit log semua mutasi.

### 3. Tech Stack & Versi

- Runtime: Node.js >= 22 · Package manager: pnpm 11.5.
- Framework: Next.js 15.5 (App Router) · React 19 · TypeScript strict.
- Database: PostgreSQL 16 + Drizzle ORM 0.45 · driver postgres 3.
- Auth: jose 5 (JWT) + bcrypt 5 · validasi input: Zod 3.
- UI: Tailwind CSS + shadcn (wrapper custom) + lucide-react · base-ui.
- Dokumen: pdf-lib 1.17 (surat jalan, berita acara, label) · QR: qrcode 1.5.
- Push: firebase-admin 14 (FCM) · ID: nanoid 5.
- Testing: Vitest (unit + integrasi) + Playwright (E2E) · CI: GitHub Actions.

### 4. Infrastruktur & Layanan Pihak Ketiga

- Hosting aplikasi: Vercel (utama) dan Coolify + Docker (jalur deploy kedua) — keduanya aktif.
- Database: Neon PostgreSQL 16 (cloud) + container PostgreSQL dev/QA lokal.
- Firebase Cloud Messaging (project "back-ohmes") — push notifikasi mobile.
- GitHub: repositori + CI/CD (lint, test, build, security audit, E2E).
- Domain + TLS: produksi (koordinasi kontrak API dengan tim mobile — domain dan error-envelope TIDAK boleh diubah sepihak).
- Catatan RAB: komponen berbiaya = hosting Vercel (plan sesuai traffic), Neon DB, server Coolify (VM), Firebase (level gratis cukup untuk volume notifikasi kecil-menengah), domain tahunan.

### 5. Inventori Fitur per Modul

- Auth & Sesi: login JWT + OTP 2 lapis, SUPERADMIN 2FA (maks 3 akun, token 5 menit), single-session mobile (409 SESSION\_EXISTS + revoke), TTL token per role, switch scope, session management.

- RBAC: 14 role (SUPERADMIN, HQ\_ADMIN, HQ\_ANALYST, HQ\_AUDITOR, AREA\_COORDINATOR, AREA\_QA, AREA\_SJ\_OFFICER, PLANT\_MANAGER, SHIFT\_SUPERVISOR, GUDANG\_INBOUND, GUDANG\_OUTBOUND, EKSPEDISI, OPERATOR\_KECER, OPERATOR\_MEMBER) + 72 permission; enforcement di API dan sidebar per role.
- Master Data: mesin (MAKER/HLP/WR/SLOP/BAL + machine template range yield), plant/region/company, produk, consumable & sparepart dengan applicable\_machines, user + assignment.
- Produksi MAKER (tablet): start shift (roster auto-pick, validasi mesin MAKER saja), buka sesi 1-6 boks TSG dari inventory FIFO (override ber-permission + audit), timbang batangan kolektif proporsional, yield server-side, batch btc\_<mesin>\_<tgl>\_<seq>, event konsumsi/downtime/maintenance per boks-sesi, waste 4 kategori wajib (MENIR/RIJEKAN/DEBU\_KASAR/DEBU\_HALUS), handoff sisa TSG antar shift (boleh beda mesin), end shift -> COMPLETED + FCM, approve -> LOCKED + CORRECTION + reopen.
- HLP (tablet): sesi HLP (auto-tutup idle 6 jam), catat packing per batch (1 batch 1 kali; total batang & berat/batang server-side), reject pack = batangan + alasan, ambang reject 5% -> push FCM, ledger rijekan 2 satuan, input operasional (material PEMAKAIAN/WASTE, downtime, maintenance mesin), rantai produksi WR -> SLOP -> BAL (batch\_stage\_event, mesin opsional/manual, stage otomatis ke tertinggi) + ringkasan sisa per stage.
- WMS Inbound: receiving TSG manual + via Surat Jalan Supplier, pool label QR (scan -> assign + jenis + berat, VOID, validasi jumlah di pabrik), reject receiving, transfer antar pabrik, retur supplier, inventory FIFO per plant.
- WMS Outbound: karton (kapasitas pack, validasi CARTON\_FULL & PACK\_INSUFFICIENT), isi pack dari batch, closeCarton, finished goods receiving (ekspektasi vs aktual dari approve).
- Dispatch & Dokumen: surat jalan dispatch PDF resmi (kop + tabel + 3 tanda tangan), Berita Acara Serah Terima & Retur PDF, dokumen material-out.
- Makloon: penerimaan batangan external (batch btx\_, source EXTERNAL) + approval/reject, entry stage (BATANGAN/PACK/PACK\_WRAPPED/SLOP/BAL), keluar ke customer dengan exit stage, PDF serah terima.
- Laporan & Dashboard: laporan TSG lengkap (masuk, stok, pakai, keluar), rekap biaya material & sparepart, laporan rijekan, dashboard per role (plant grafik / area harian-mingguan + perbandingan antar pabrik + region picker / HQ grafik), KPI.
- Label & Cetak: label pool 100x75mm multi-halaman (inisial 43pt + kotak angka + QR asli backend), cetak label standalone, cetak roster bulanan, surat jalan supplier.
- Notifikasi FCM: shift COMPLETED (PM + supervisor), receiving PENDING, reject HLP tinggi, batangan external PENDING; payload data untuk deep-link mobile (type + id rute); token mati auto-bersih.

## 6. Basis Data

- 66 tabel PostgreSQL 16, 34 migrasi (journal drizzle + migrasi manual idempotent yang auto-apply tiap deploy).
- Row Level Security aktif via role mes\_app (non-superuser) — semua tabel operasional wajib plantId.
- Konvensi: soft delete (deletedAt), audit log semua mutasi, idempotency key untuk operasi idempotent.
- Seed idempotent + sync permission-role additive (grant permission baru sampai ke produksi tanpa migrasi).

## 7. Keamanan

- Autentikasi 2 lapis (password + OTP), 2FA SUPERADMIN, OTP bypass terkontrol.
- Otorisasi ganda: RBAC middleware tiap route + RLS PostgreSQL (pertahanan berlapis).
- Rate limiting per path (middleware), CORS ketat.
- Audit trail seluruh mutasi (who, what, when, before/after).
- Sesi: revoke paksa, single-session mobile, TTL per role (lantai produksi 8 jam).
- API error-envelope standar + requestId (koordinasi kontrak mobile).

## 8. Perangkat Lapangan & Integrasi

- Tablet produksi: web responsive (/tablet — start-shift, shift aktif, HLP, gudang, label).
- Mobile app (Flutter): login, approve shift, receiving, dashboard, FCM push + deep-link.

- Printer label: XPrinter 420B (label 100x75mm, PDF multi-halaman).
- QR: label asli dari backend (qrcode), scan via kamera tablet/HP.

## 9. Pengujian & Kualitas

- Unit test: 122 · Integrasi: 22 · E2E Playwright: 10 spec rantai bisnis penuh (produksi -> approval -> area -> HQ, HLP, gudang inbound/outbound, dispatch, transfer/retur).
- CI GitHub Actions 5 job: lint, test, build, security audit, E2E (reset DB + build + server dedicated tiap run).
- Audit dependency rutin (1 vuln sedang ditoleransi: uuid transitif firebase-admin).
- Coverage line 9% (target aspirasional 80%).

## 10. Operasional

- Deploy: Vercel + Coolify (entrypoint otomatis: migrate + migrasi manual + build).
- Backup & recovery: pg\_dump runbook, dokumen runbook operasional (docs/17, docs/18).
- Auto-cleanup: sesi produksi 24 jam (instrumentation).
- Utilitas: reset data transaksi (scripts/reset-transactions.sql), test push FCM (scripts/fcm-test-push.mjs), generator dokumen PDF.

## 11. Catatan untuk Penyusunan RAB

- Biaya berulang (bulanan/tahunan): hosting Vercel, Neon PostgreSQL, VM Coolify, domain + TLS, Firebase (perkiraan gratis - kecil).
- Perangkat per pabrik: tablet rantai produksi, HP Android operator/approver, printer label XPrinter 420B, koneksi internet.
- Pengembangan tersisa (estimasi): checklist go-live lapangan (re-login user, verifikasi FCM natural, checklist manual), coverage test aspirasional 80%, halaman mobile baru bila diminta (external receiving, batch detail), peringatan selisih stage (docs/25 - belum diimplementasikan).
- Dukungan & maintenance: pemeliharaan rutin, monitoring deploy, koordinasi kontrak API dengan tim mobile.
- Dokumentasi tersedia: PRD (docs/01), API spec (docs/06), RBAC matrix (docs/05), runbook operasional (docs/17), deploy Coolify (docs/deploy-coolify-runbook.md), PDF alur produksi dan skenario pengujian HLP.